

Chapter 9

Kinematics of Rigid Bodies I

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MC2103
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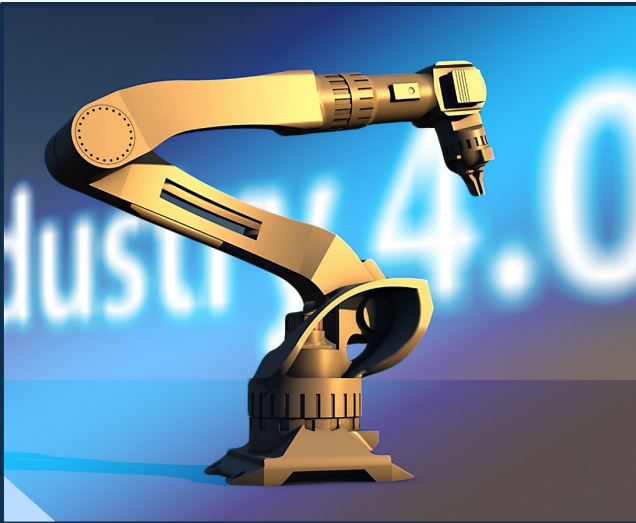
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9 Kinematics of Rigid Bodies I

Introduction

In Week 2, we studied Kinematics of particles. However, the same principle may not be applied the Kinematics of rigid bodies, specifically in three dimensional spaces.

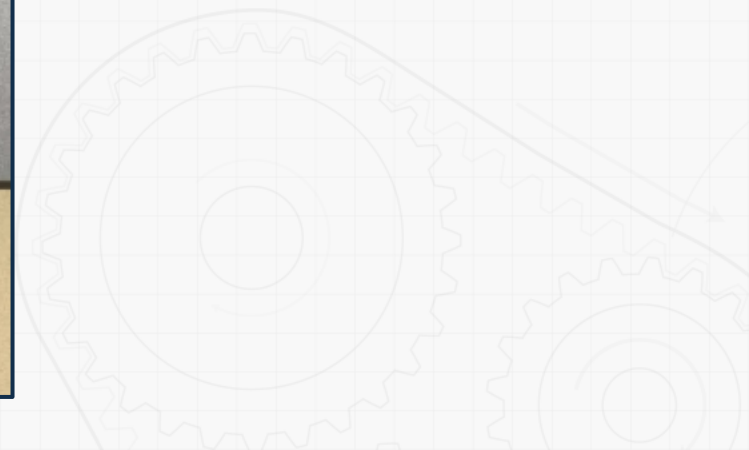
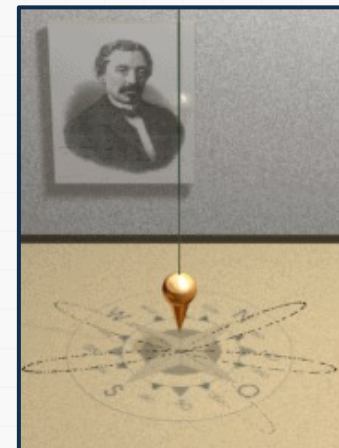
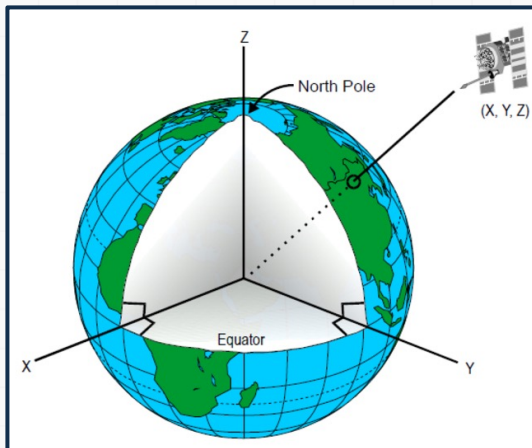


9 Kinematics of Rigid Bodies I

Reference Frame

⚙ References

- Same as the rigid body of interest without considering mass and volume
- Usually represented by capital alphabet (e.g., A, D)
- Inertial reference frame is a reference frame where all points have zero acceleration.

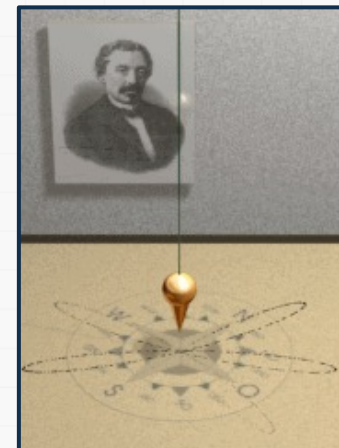
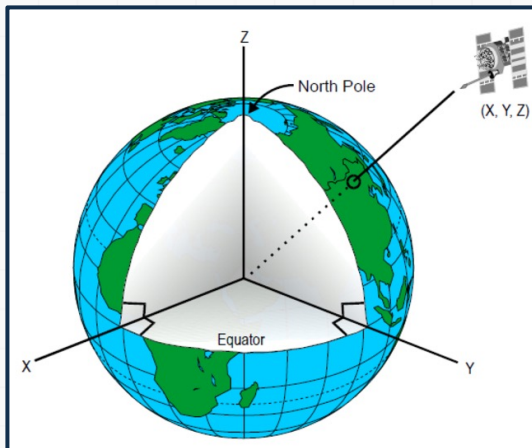


9 Kinematics of Rigid Bodies I

Reference Frame

References

- Acceleration in Newton's law of motion must be observed in an inertial reference frame.
- Earth is not an ideal inertial reference frame (e.g., any rotations involve acceleration).
- Inertial reference frame is also called Newtonian reference frame or absolute reference frame.



9 Kinematics of Rigid Bodies I

Coordinate System

⚙️ Position Vector

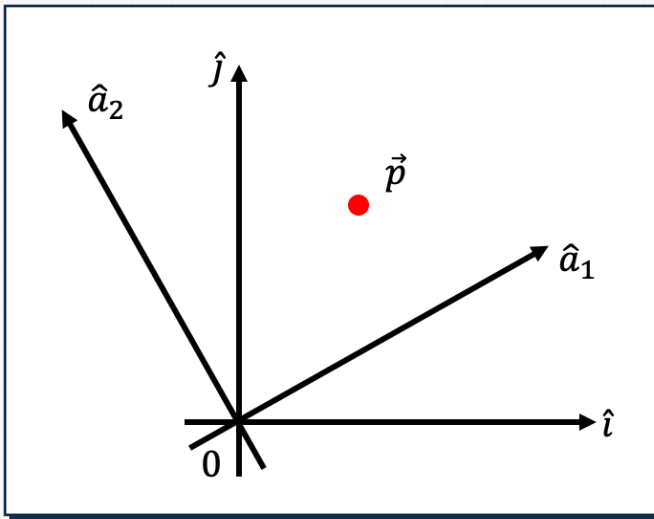
- A position vector is a geometric quantity that does not require any specific numbers.
- However, once a coordinate system is defined, the position vector can be assigned specific numbers, called a coordinate.
- In 3D space, a coordinate system has three unit vectors that are dextral and orthogonal.

ex: $(\hat{i}, \hat{j}, \hat{k})$, $(\hat{e}_t, \hat{e}_n, \hat{e}_b)$, $(\hat{e}_r, \hat{e}_\theta, \hat{k})$, $(\hat{a}_1, \hat{a}_2, \hat{a}_3)$

9 Kinematics of Rigid Bodies I

Coordinate System

⚙️ Position Vector



- Multiple coordinate systems can be assigned to a reference frame.
- One position vector can have different coordinates if different coordinate systems are used.

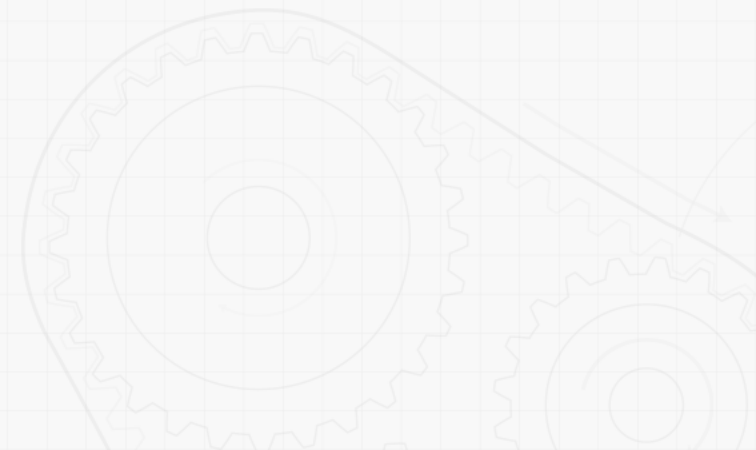
$$\begin{aligned}\vec{p} &= p_x \hat{i} + p_y \hat{j} + p_z \hat{k} = p_1 \hat{a}_1 + p_2 \hat{a}_2 + p_3 \hat{a}_3 = \dots \\ &= (p_x, p_y, p_z)_{ijk} = (p_1, p_2, p_3)_{a_1 a_2 a_3} = \dots\end{aligned}$$

9 Kinematics of Rigid Bodies I

Coordinate

⚙ Coordinates

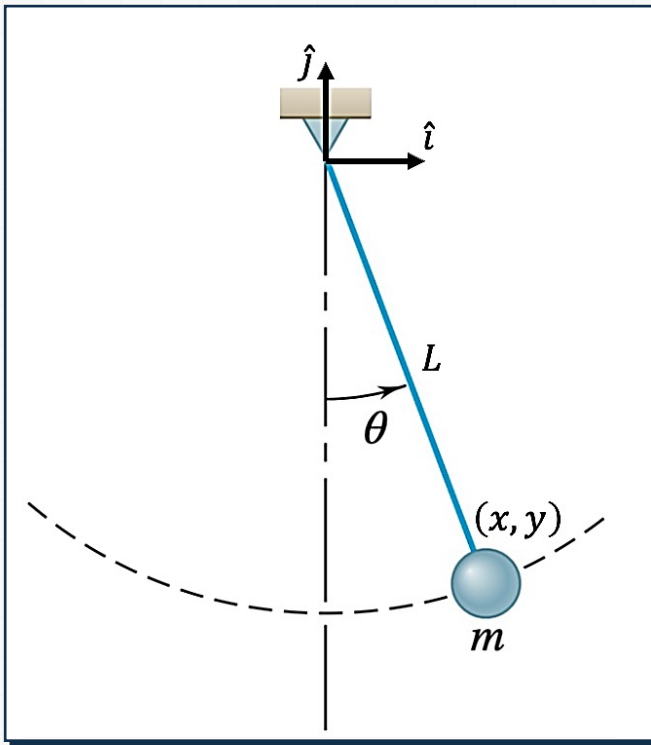
- In the previous page, we learned that coordinates can represent a position once a coordinate system is defined.
- In general, coordinates are scalar variables that determine the **configuration** of a system.
- The minimum number of coordinates is usually called degrees of freedom (DOFs).
- Coordinates with minimum number are (usually, but not necessarily) called generalized coordinates.



9 Kinematics of Rigid Bodies I

Coordinate

⚙ Coordinates



- What scalar variables do you need to describe the tip of the pendulum with a length L ?

$$\begin{aligned}\vec{p} &= x\hat{i} + y\hat{j} \\ &= L \sin \theta \hat{i} - L \cos \theta \hat{j}\end{aligned}$$

- What are the scalar variables and how many?

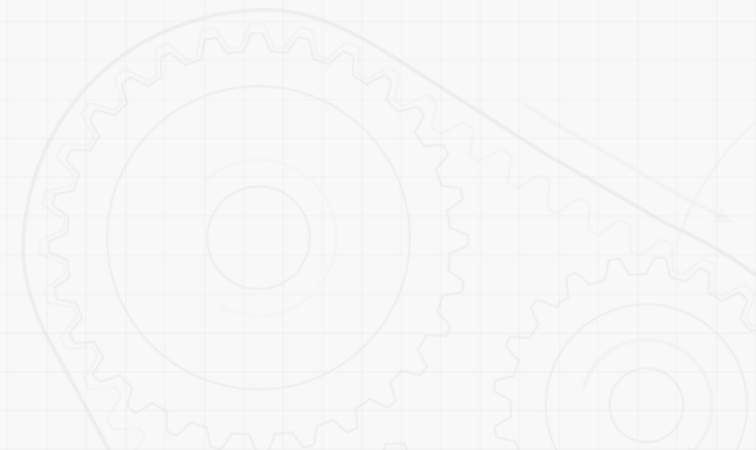
Generalized coordinates?

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Virtual Reference Frame

⚙ Define Virtual Reference Frame

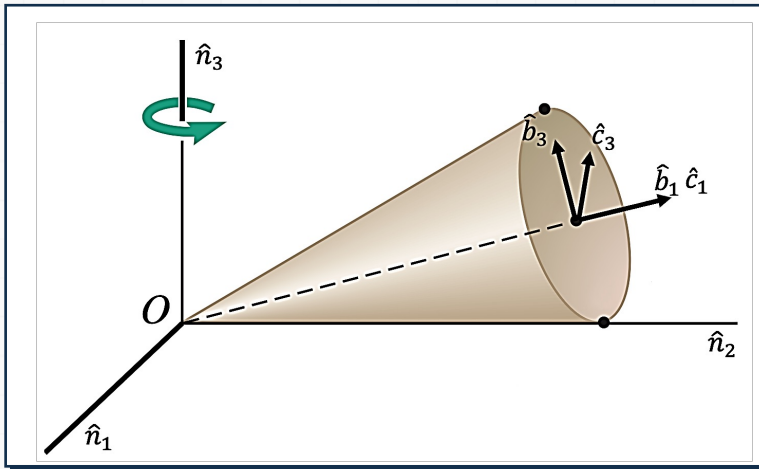
- Sometimes, it is useful to define a virtual reference frame that does not align with a rigid body.
- Introducing a virtual reference frame may make it easier to analyze the motion of the rigid body.



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Virtual Reference Frame

⚙️ Identify appropriate reference frames



- The example shows the virtual reference frame.

Identify appropriate reference frames among $\hat{n}_1\hat{n}_2\hat{n}_3$, $\hat{b}_1\hat{b}_2\hat{b}_3$, $\hat{c}_1\hat{c}_2\hat{c}_3$

- (1) Newtonian reference frame:
- (2) Body reference frame:
- (3) Virtual reference frame:

9 Kinematics of Rigid Bodies I

Properties of Vectors

⚙ Properties

- Vectors need to satisfy the following properties (among 7-8 axioms).

$$\vec{x} + \vec{y} = \vec{y} + \vec{x}$$

Commutativity

$$(\vec{x} + \vec{y}) + \vec{z} = \vec{x} + (\vec{y} + \vec{z})$$

Associativity

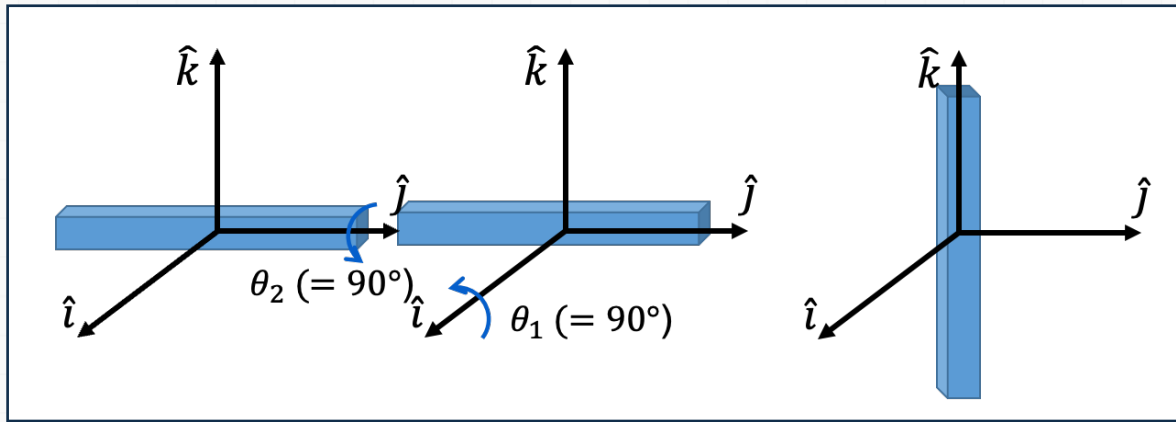
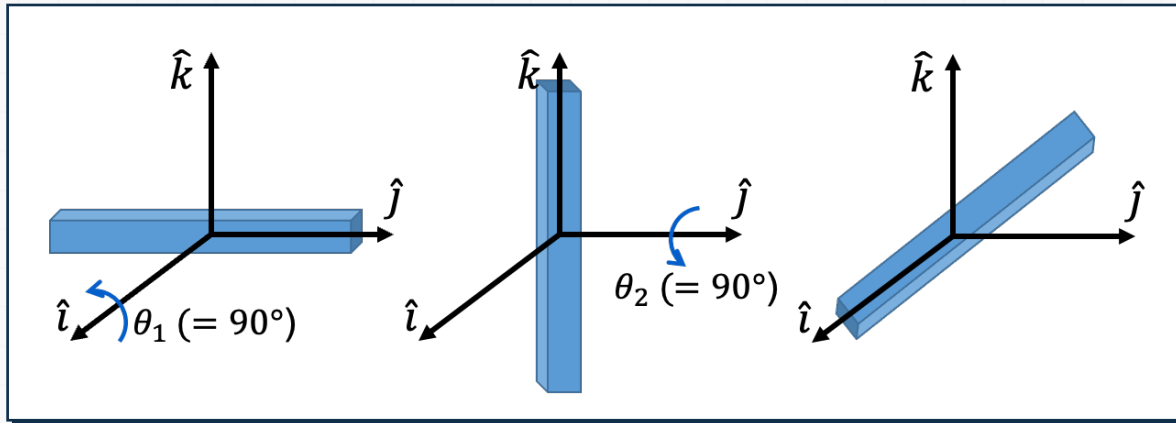
$$\alpha(\vec{x} + \vec{y}) = \alpha\vec{x} + \alpha\vec{y}$$

Distributivity

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Orientations are not Vectors

⚙ Two consecutive rotations



- Consider two consecutive rotations: one about $\hat{i}$ axis by angle θ_1 and another one about $\hat{j}$ axis by angle θ_2
- Since the following commutativity does not hold,

$$\theta_1 \hat{i} + \theta_2 \hat{j} \neq \theta_2 \hat{j} + \theta_1 \hat{i}$$

orientations are not vectors!

9 Kinematics of Rigid Bodies I

Vector Differentiation in Time

⚙️ Differentiation

- Assume that a vector $\vec{p}(t)$ is observed in reference frame A .
Also, assume that a coordinate system $\hat{a}_1\hat{a}_2\hat{a}_3$ is attached to the reference frame A .

Then, $\vec{p}(t)$ can be expressed as follows:

$$\vec{p}(t) = p_1(t)\hat{a}_1 + p_2(t)\hat{a}_2 + p_3(t)\hat{a}_3$$

or

$$\vec{p} = p_1\hat{a}_1 + p_2\hat{a}_2 + p_3\hat{a}_3$$

9 Kinematics of Rigid Bodies I

Vector Differentiation in Time

⚙️ Differentiation

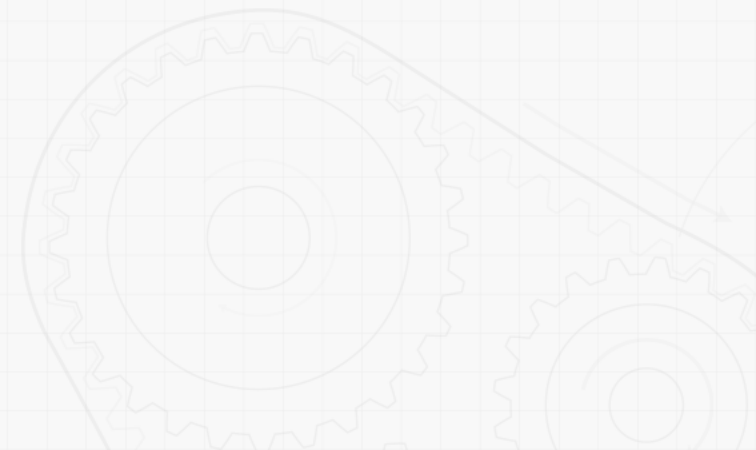
- If an observer (e.g., a person) in reference frame A sees that $\vec{p}$ changes over time, its rate change is defined as follows:

$${}^A \frac{d\vec{p}}{dt} = \frac{dp_1}{dt} \hat{a}_1 + \frac{dp_2}{dt} \hat{a}_2 + \frac{dp_3}{dt} \hat{a}_3$$

Vector Differentiation in Time

⚙️ Differentiation

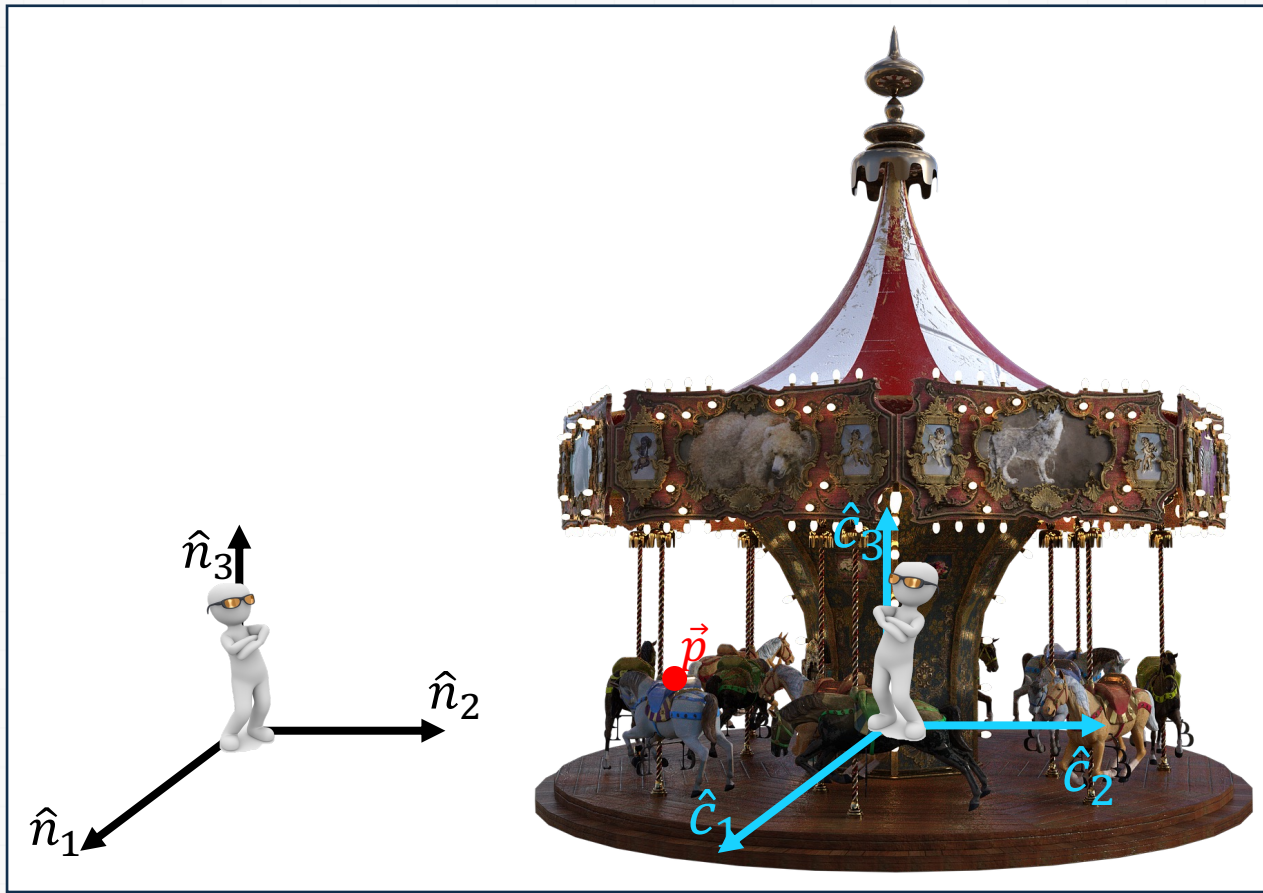
- It is important to note that vector differentiation in time depends on what reference frame the observer resides in.
- That is, a rate change of $\vec{p}$ observed from reference frame A is possibly different from a rate change of $\vec{p}$ observed from reference frame B .
ex) Carousel (merry-go-round)



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Vector Differentiation in Time: Example

⚙️ Carousel (merry-go-round)



- Two observers:
one in N and one in C .
- Two observers see the same
point $\vec{p}$.
- Two observers see
different motion:
Observer in C : vertical movement
Observer in N : vertical
movement, rotation, translation

Notation of Motion Vectors

⚙️ Carousel (merry-go-round)

- As we have noticed, a motion of one position, $\vec{p}$, may be different depending on where the observers are.
- Motion vectors: velocity and acceleration
- Therefore, it is important to indicate the reference frame when differentiating vectors.
- Left superscript: reference frame of the observer,
Right superscript: point of interest

Notation of Motion Vectors

⚙ Motion Vectors

- As we have noticed, a motion of one position, $\vec{p}$, may be different depending on where the observers are.
- Motion vectors: velocity and acceleration
- Therefore, it is important to indicate the reference frame when differentiating vectors.

Left superscript

Reference frame of
the observer

Right superscript

Point of interest

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Notation of Motion Vectors

⚙️ Velocity and Acceleration

• Velocity

$$\begin{aligned} {}^A\vec{v}^P &= \frac{{}^A d\vec{p}}{dt} = \frac{dp_{a1}}{dt} \hat{a}_1 + \frac{dp_{a2}}{dt} \hat{a}_2 + \frac{dp_{a3}}{dt} \hat{a}_3 \\ {}^B\vec{v}^P &= \frac{{}^B d\vec{p}}{dt} = \frac{dp_{b1}}{dt} \hat{b}_1 + \frac{dp_{b2}}{dt} \hat{b}_2 + \frac{dp_{b3}}{dt} \hat{b}_3 \end{aligned}$$

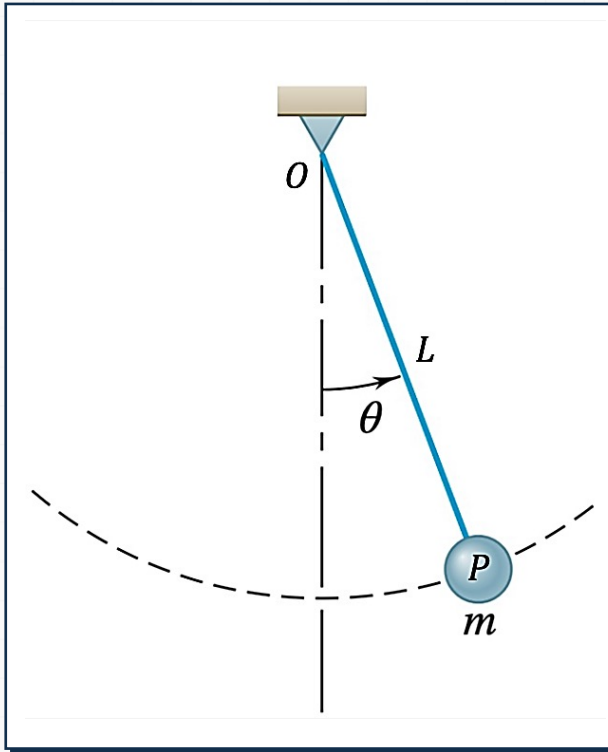
• Acceleration

$${}^A\vec{a}^P = \frac{{}^A d}{dt} ({}^A\vec{v}^P)$$

9 Kinematics of Rigid Bodies I

Sample Problem 1

⚙ Consider the following pendulum problem.



• Determine

- Velocity of point p observed at N
- Velocity of point p observed at A

9 Kinematics of Rigid Bodies I

Sample Problem 1

⚙️ **Consider the following pendulum problem.**

• **Strategy**

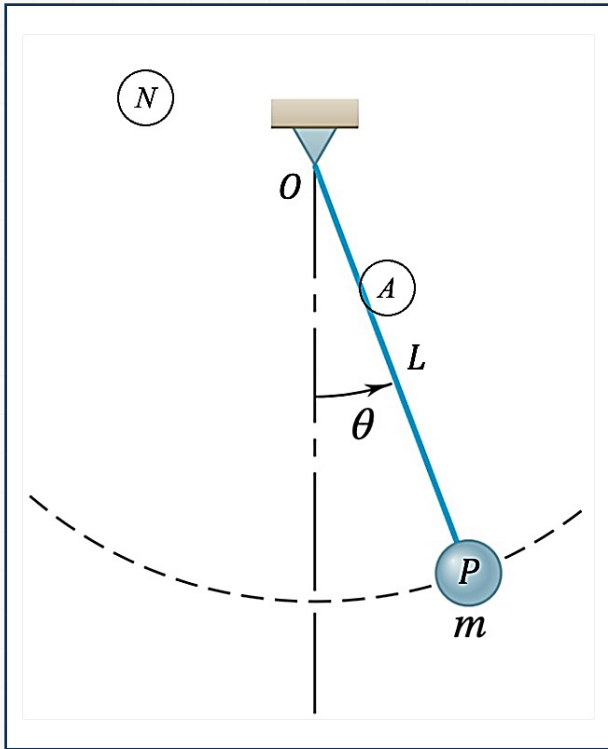
- Identify reference frames: inertial reference frame, body reference frame.
- Assign coordinate systems to each reference frame.
- Express the position vector $\vec{p}$ using coordinate systems and the corresponding coordinates.
- Perform differentiation in terms of each reference frame

9 Kinematics of Rigid Bodies I

Sample Problem 1

⚙ Identify reference frames

- Each rigid body has one reference frame



Inertial reference frame: N

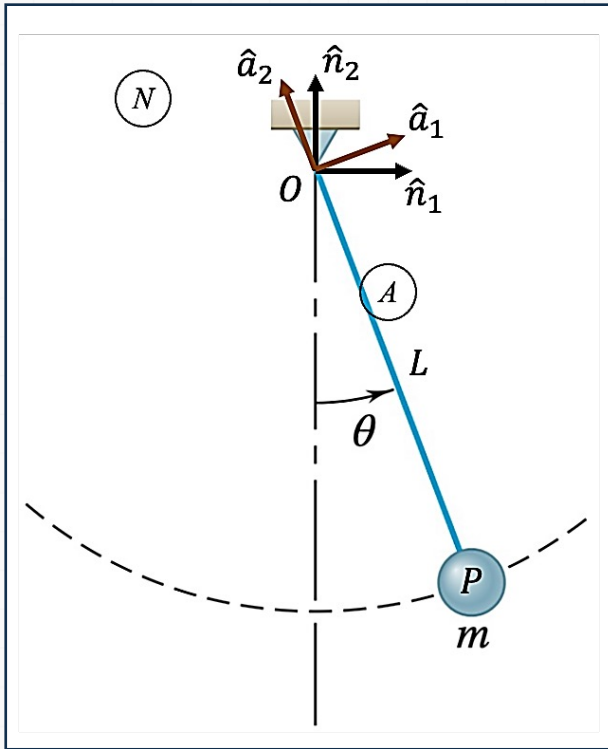
Body reference frame of pendulum: A

9 Kinematics of Rigid Bodies I

Sample Problem 1

⚙️ Assign a coordinate system to each reference frame

- Each reference frame has at least one coordinate system



Inertial reference frame: N
Assign $O\hat{n}_1\hat{n}_2\hat{n}_3$

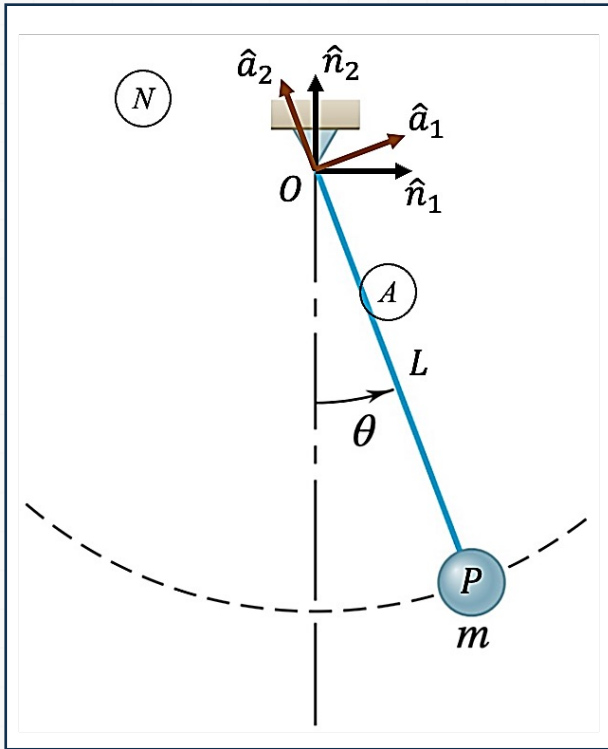
Body reference frame of pendulum: A
Assign $O\hat{a}_1\hat{a}_2\hat{a}_3$

9 Kinematics of Rigid Bodies I

Sample Problem 1

⚙ Express the position vector $\vec{p}$

- Position vector can be expressed in terms of all coordinate systems



In terms of $O\hat{n}_1\hat{n}_2\hat{n}_3$

$$\vec{p} = \overrightarrow{OP} = L \sin \theta \hat{n}_1 - L \cos \theta \hat{n}_2$$

In terms of $O\hat{a}_1\hat{a}_2\hat{a}_3$

$$\vec{p} = \overrightarrow{OP} = -L\hat{a}_2$$

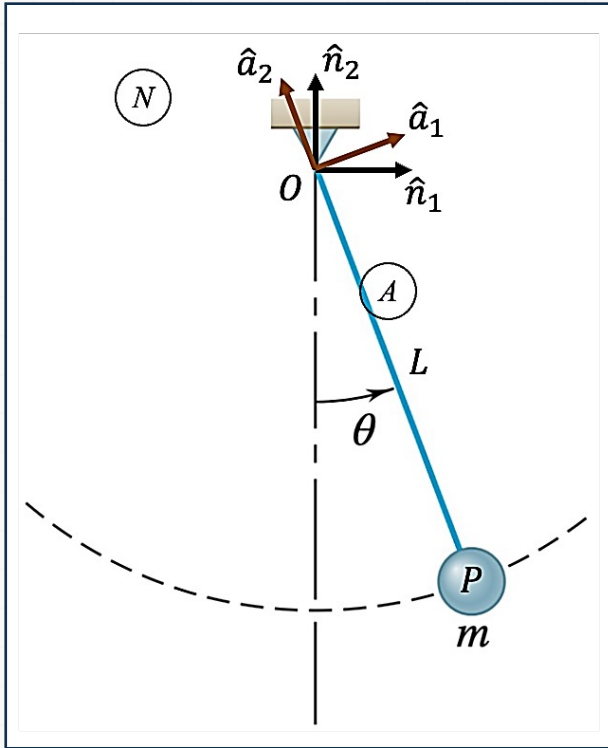
Note that both expressions are identical

9 Kinematics of Rigid Bodies I

Sample Problem 1

⚙ Differentiate $\vec{p}$ across reference frames

- Integrate twice to find $v(t)$ and $y(t)$



In terms of reference frame N

$$\begin{aligned} {}^N\vec{v}^P &= \frac{{}^N d\vec{p}}{dt} = \frac{dp_{n1}}{dt} \hat{n}_1 + \frac{dp_{n2}}{dt} \hat{n}_2 \\ &= L\dot{\theta} \cos \theta \hat{n}_1 + L\dot{\theta} \sin \theta \hat{n}_2 \end{aligned}$$

In terms of reference frame A

$$\begin{aligned} {}^A\vec{v}^P &= \frac{{}^A d\vec{p}}{dt} = \frac{dp_{a1}}{dt} \hat{a}_1 + \frac{dp_{a2}}{dt} \hat{a}_2 \\ &= 0 \end{aligned}$$

Note that both expressions are DIFFERENT!

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Sample Problem 1

⚙️ Reflect and Think

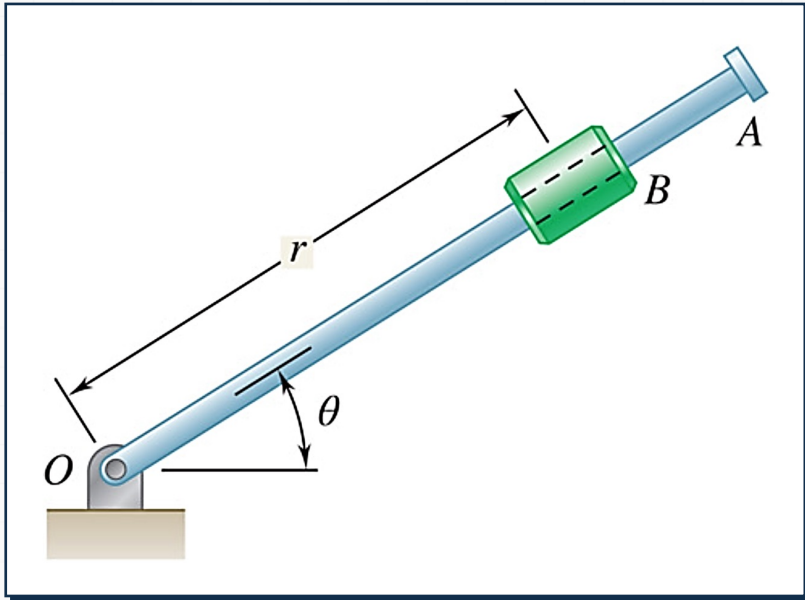
Position vectors are identical regardless of reference frames whereas motion vectors (i.e., derivatives of position vectors) differ depending on reference frames.

Differentiation of scalar variables has nothing to do with reference frames.

9 Kinematics of Rigid Bodies I

Sample Problem 2

⚙ Consider the following rotating arm with a slider



• Determine

- Velocity of slider B observed at N
- Acceleration of slider B observed at N
- Velocity of slider B observed at A
- Acceleration of slider B observed at A

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Sample Problem 2

⚙️ **Consider the following rotating arm with a slider**

• **Strategy**

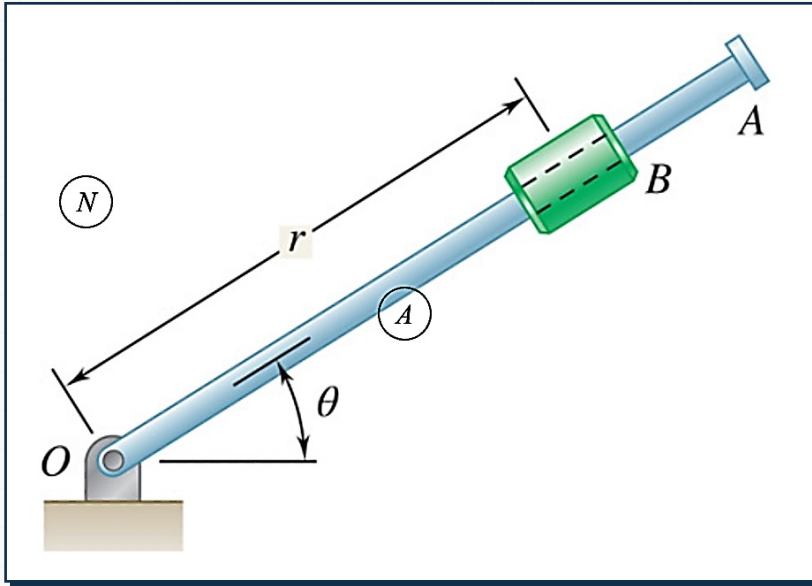
- Identify reference frames: inertial reference frame, body reference frame.
- Assign coordinate systems to each reference frame.
- Express the position vector $\vec{p}$ of the slider B using coordinate systems and the corresponding coordinates.
- Perform differentiation in terms of each reference frame

9 Kinematics of Rigid Bodies I

Sample Problem 2

⚙ Identify reference frames

- Each rigid body has one reference frame



Inertial reference frame: N

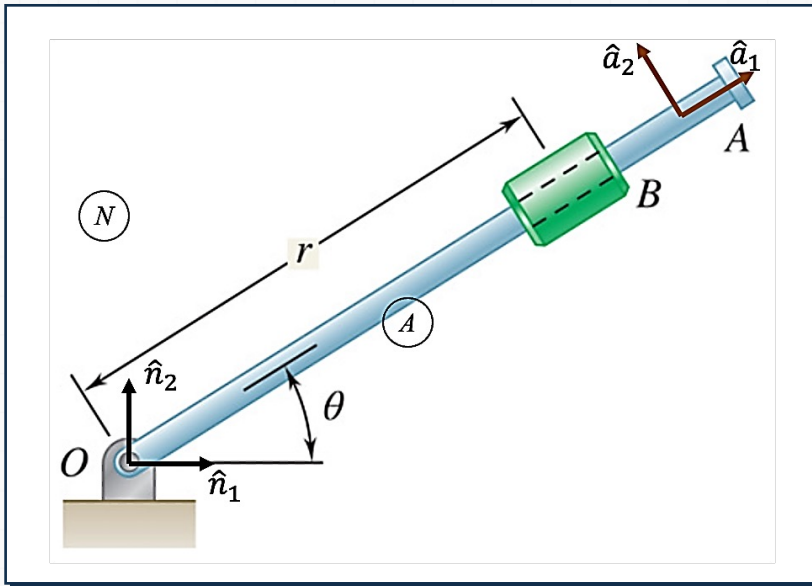
Body reference frame of arm: A

9 Kinematics of Rigid Bodies I

Sample Problem 2

⚙ Assign a coordinate system to each reference frame

- Each reference frame has at least one coordinate system



Inertial reference frame: N

Assign $O\hat{n}_1\hat{n}_2\hat{n}_3$

Body reference frame of pendulum: A

Assign $O\hat{a}_1\hat{a}_2\hat{a}_3$

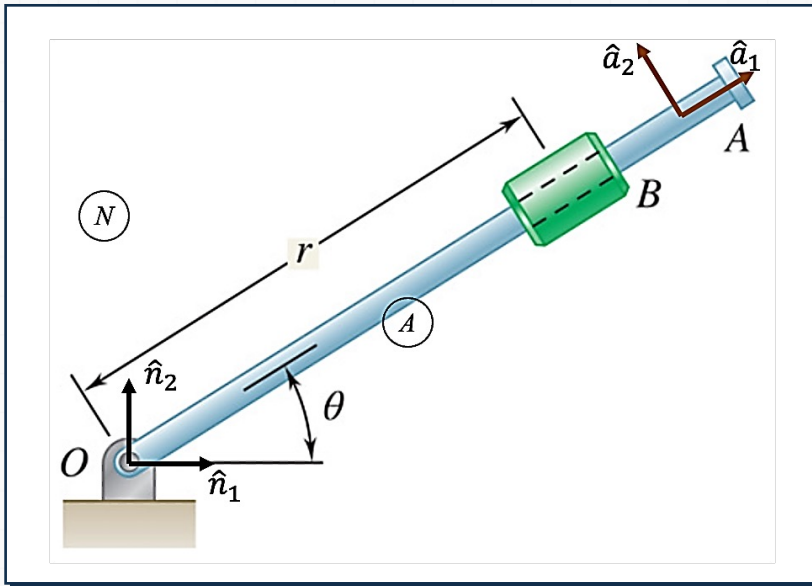
Why are $\hat{a}_1\hat{a}_2\hat{a}_3$ axes not at O ?

9 Kinematics of Rigid Bodies I

Sample Problem 2

⚙ Express the position vector $\vec{p}$

- Position vector can be expressed in terms of all coordinate systems



In terms of $O\hat{n}_1\hat{n}_2\hat{n}_3$

$$\vec{p} = \overrightarrow{OB} = r \cos \theta \hat{n}_1 + r \sin \theta \hat{n}_2$$

In terms of $O\hat{a}_1\hat{a}_2\hat{a}_3$

$$\vec{p} = \overrightarrow{OP} = r \hat{a}_1$$

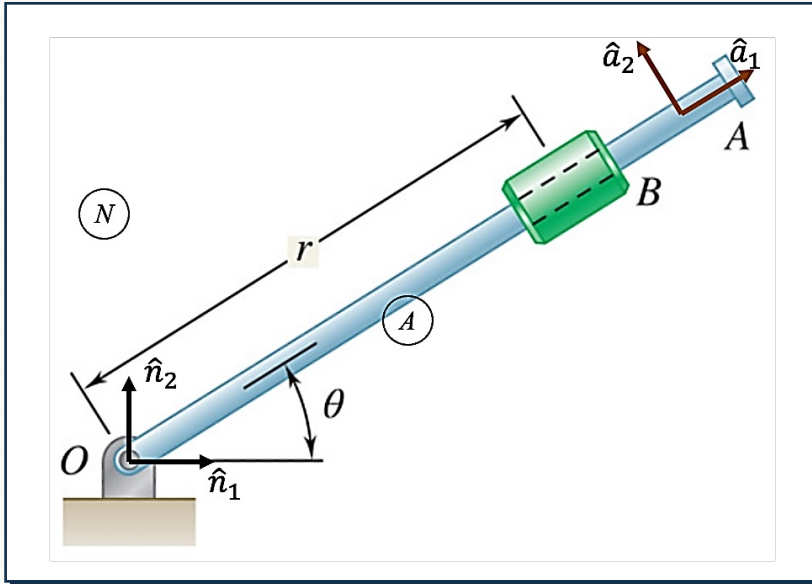
Note that both expressions are identical

9 Kinematics of Rigid Bodies I

Sample Problem 2

⚙ Differentiate $\vec{p}$ across reference frames

- Differentiate $\vec{p}$ with respect to reference frames to find ${}^N\vec{v}^P$ and ${}^A\vec{v}^P$



In terms of reference frame N

$$\begin{aligned} {}^N\vec{v}^P &= \frac{{}^N d\vec{p}}{dt} = \frac{dp_{n1}}{dt} \hat{n}_1 + \frac{dp_{n2}}{dt} \hat{n}_2 \\ &= (\dot{r} \cos \theta - r\dot{\theta} \sin \theta) \hat{n}_1 + (\dot{r} \sin \theta + r\dot{\theta} \cos \theta) \hat{n}_2 \end{aligned}$$

In terms of reference frame A

$$\begin{aligned} {}^A\vec{v}^P &= \frac{{}^A d\vec{p}}{dt} = \frac{dp_{a1}}{dt} \hat{a}_1 + \frac{dp_{a2}}{dt} \hat{a}_2 \\ &= \dot{r} \hat{a}_1 \end{aligned}$$

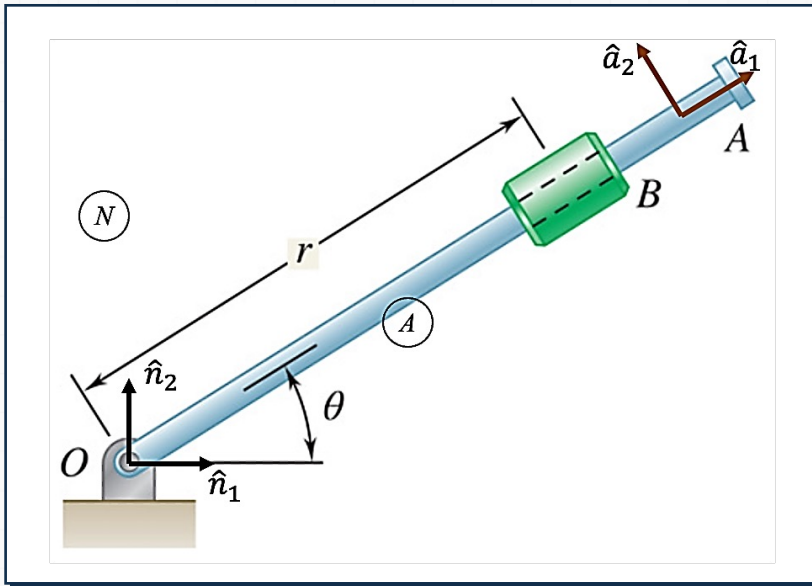
Note that both expressions are DIFFERENT!

9 Kinematics of Rigid Bodies I

Sample Problem 2

⚙️ Differentiate $\vec{v}$ across reference frames

- Differentiate $\vec{v}$ with respect to reference frames to find ${}^N\vec{a}^P$ and ${}^A\vec{a}^P$



In terms of reference frame N

$$\begin{aligned} {}^N\vec{a}^P &= \frac{d}{dt}({}^N\vec{v}^P) \\ &= (\ddot{r} \cos \theta - 2\dot{r}\dot{\theta} \sin \theta - r\ddot{\theta} \sin \theta - r\dot{\theta}^2 \cos \theta)\hat{n}_1 \\ &\quad + (\ddot{r} \sin \theta + 2\dot{r}\dot{\theta} \cos \theta + r\ddot{\theta} \cos \theta - r\dot{\theta}^2 \sin \theta)\hat{n}_2 \end{aligned}$$

In terms of reference frame A

$${}^A\vec{a}^P = \frac{d}{dt}({}^A\vec{v}^P) = \ddot{r}\hat{a}_1$$

Note that both expressions are DIFFERENT!

9 Kinematics of Rigid Bodies I

Sample Problem 2

⚙️ Reflect and Think

Velocity and acceleration vectors are much simpler with body reference frame. This is because the observer in the body reference frame does not see the movement of the body itself. The following is also possible:

$$\vec{p} = \overrightarrow{OP} = r\hat{a}_1$$

$${}^N\vec{v}^P = \frac{{}^N d\vec{p}}{dt} = \dot{r}\hat{a}_1 + r\dot{\hat{a}}_1 = \dot{r}\hat{a}_1 + r\dot{\theta}\hat{a}_2$$

$${}^N\vec{a}^P = \frac{{}^N d}{dt}({}^N\vec{v}^P) = \ddot{r}\hat{a}_1 + \dot{r}\dot{\hat{a}}_1 + \dot{r}\dot{\theta}\hat{a}_2 + r\ddot{\theta}\hat{a}_2 + r\dot{\theta}\dot{\hat{a}}_2 = (\ddot{r} - r\dot{\theta}^2)\hat{a}_1 + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{a}_2$$

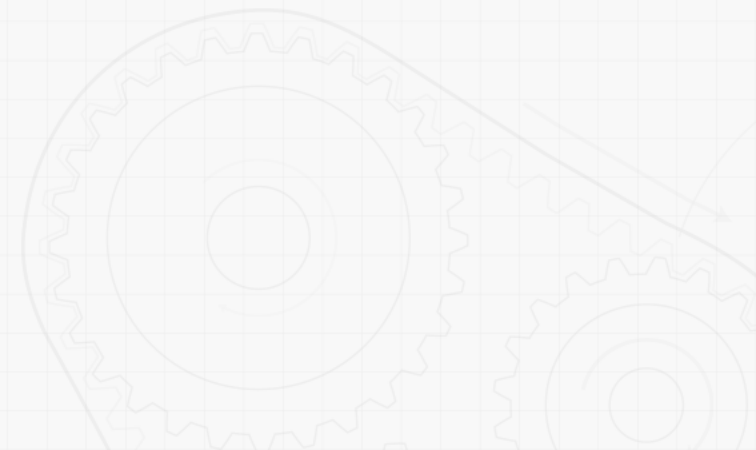
9 Kinematics of Rigid Bodies I

Sample Problem 3

⚙️ **Show that orientation is not a vector using commutativity**

• **Strategy**

- Find a counterexample where vector addition does not hold commutativity.

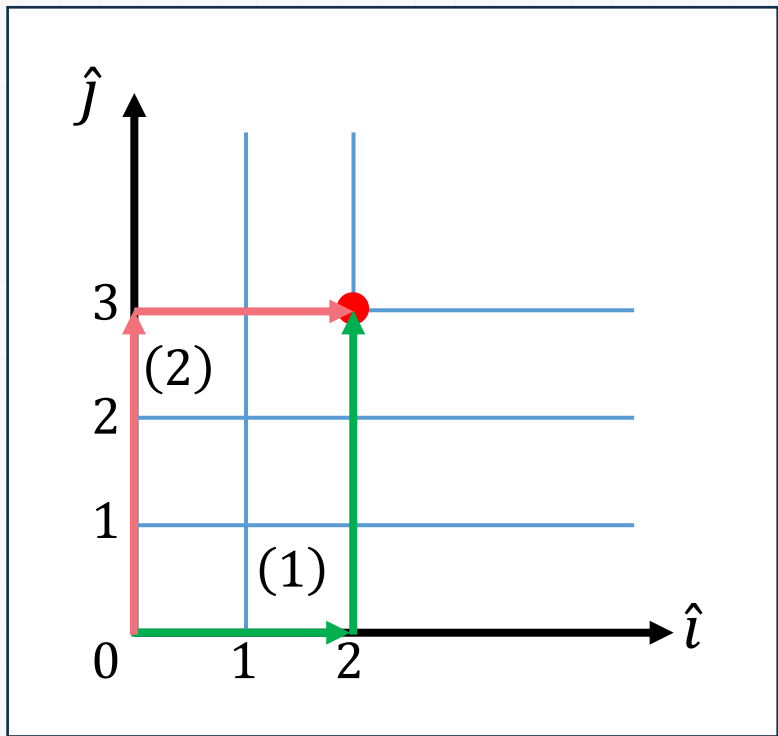


9 Kinematics of Rigid Bodies I

Sample Problem 3

⚙️ Check commutativity of position vectors

- Check $\vec{p} = \vec{a} + \vec{b} = \vec{b} + \vec{a}$ and find the meaning of the operation.



$$\begin{aligned}\vec{p} &= 2\hat{i} + 3\hat{j} & \dots (1) \\ &= 3\hat{j} + 2\hat{i} & \dots (2)\end{aligned}$$

Meaning of (1): move by 2 in $\hat{i}$ and then move by 3 in $\hat{j}$

Meaning of (2): move by 3 in $\hat{j}$ and then move by 2 in $\hat{i}$

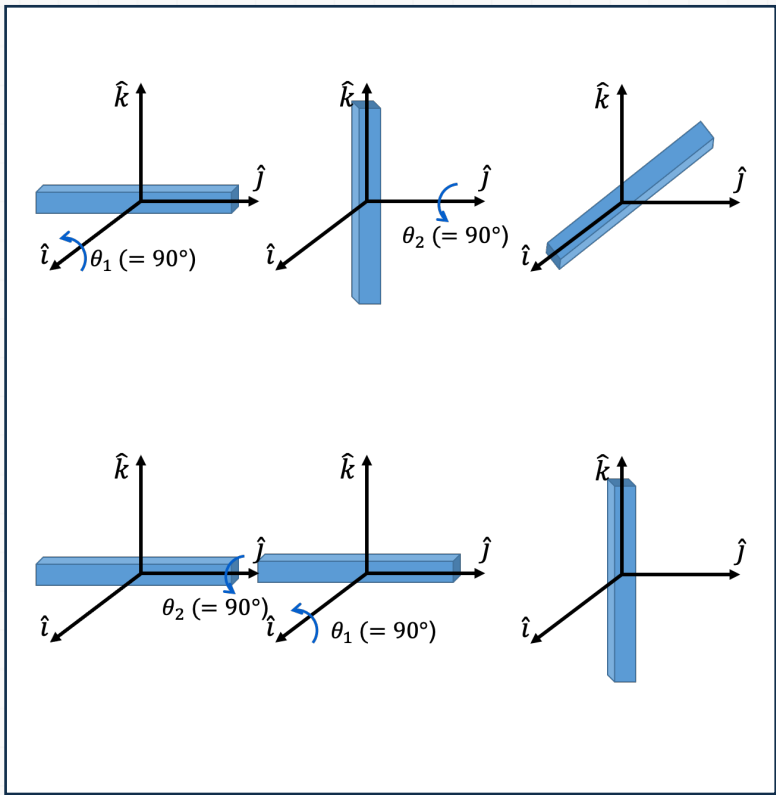
Both results are the same.

9 Kinematics of Rigid Bodies I

Sample Problem 3

⚙ Check commutativity of orientation vectors

- Check $\vec{p} = \vec{a} + \vec{b} = \vec{b} + \vec{a}$ and find the meaning of the operation.



$$\vec{p} = 90^\circ \hat{i} + 90^\circ \hat{j} \quad \dots (1)$$

$$= 90^\circ \hat{j} + 90^\circ \hat{i} \quad \dots (2)$$

Meaning of (1): rotate by 90° in $\hat{i}$ and then rotate by 90° in $\hat{j}$

Meaning of (2): rotate by 90° in $\hat{j}$ and then rotate by 90° in $\hat{i}$

Both results are different.
Therefore, orientation is not a vector.

Sample Problem 3

⚙️ Reflect and Think

We checked that orientation does not satisfy commutativity and thus is not a vector.

Therefore, orientation cannot be differentiated in the way the position vector was differentiated to get velocity and acceleration. In the following lecture, we will study how to compute angular velocity.

FYI, angular velocity is a vector. Therefore, angular acceleration can be computed by differentiating the angular velocity.

References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, McGraw Hill, Chapter 15
- 유홍희, "동역학", 1판, 문운당, 8장